The inventor of basic drilling machine is Arthur James Arnot and William Blanch Brain of Melborne. Australia by using the electric motor, in the year of 1889.


Arthur James Arnot

2.1 Introduction

The Drilling Machine is one of the most important machine tools in a workshop. As regards its importance it is second only to the lathe. It was designed to produce a cylindrical hole of required diameter and depth on work pieces.

Though holes can be made by different machine tools in a shop, drilling machine is designed specifically to perform the operation of drilling and similar operations. Drilling can be done easily at a low cost in a Shaffer period of time in a drilling machine.

Drilling can be called as the operation of producing a cylindrical hole of required diameter and depth by removing metal by the rotating the cutting edges of a drill bit.

2.2 Method of Drilling

The cutting tool known as drill bit is fitted into the spindle of the drilling machine. A mark of indentation is made at the required location with a center punch. The rotating drill is pressed at the location and is fed into the work. The hole can be made up to a required depth.

2.3 Structure of a drilling machine

The basic parts of a drilling machine are a base, column, drill head and spindle.


3D Method of Drilling

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The base made of cast iron may rest on a bench, pedestal or floor depending upon the design. Larger and heavy duty machines are grounded on the floor. The column is mounted vertically upon the base. It is accurately machined and the table can be moved up and down on it. The drill spindle an electric motor and the mechanism meant for driving the spindle at different speeds are mounted on the top of the column. Power is transmitted from the electric motor to the spindle through a flat belt (or) a 'V' belt.

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